

Apache Spark Cluster Setup on Ubuntu VMs

A Step-by-Step Installation & Configuration Guide

Master & Worker Nodes Configuration

Tutorial: Setting up Apache Spark Cluster on Ubuntu VMs

This guide provides a step-by-step procedure to install and configure an Apache Spark cluster consisting of one **Master** and multiple **Workers** on Ubuntu virtual machines.

Prerequisites

Perform these steps on **all** nodes (Master and Workers).

1. Update System

```
sudo apt update && sudo apt upgrade -y
```

2. Install Java (OpenJDK 11 or 17)

Spark requires Java. Install OpenJDK:

```
sudo apt install openjdk-11-jdk -y
# Verify installation
java -version
```

3. Configure SSH (Passwordless Login)

To allow the Master to start/stop workers, configure passwordless SSH.

```
# Install SSH server if not present
sudo apt install openssh-server -y

# Generate SSH keys (on Master only)
ssh-keygen -t rsa -P "" -f ~/.ssh/id_rsa

# Copy the key to all nodes (including itself)
ssh-copy-id user@master-ip
ssh-copy-id user@worker1-ip
ssh-copy-id user@worker2-ip
```

Spark Installation

Perform these steps on **all** nodes.

4. Download Apache Spark

Visit the Spark Downloads page (<https://spark.apache.org/downloads.html>) to get the latest version.

```
wget https://archive.apache.org/dist/spark/spark-3.5.0/spark-3.5.0-bin-hadoop3.tgz
```

5. Extract and Move

```
tar -xvzf spark-3.5.0-bin-hadoop3.tgz
sudo mv spark-3.5.0-bin-hadoop3 /opt/spark
```

6. Set Environment Variables

Add Spark to your PATH. Edit ~/.bashrc:

```
nano ~/.bashrc
```

Add the following lines at the end:

```
export SPARK_HOME=/opt/spark
export PATH=$PATH:$SPARK_HOME/bin:$SPARK_HOME/sbin
export PYSPARK_PYTHON=python3
```

Apply changes:

```
source ~/.bashrc
```

Cluster Configuration

7. Configure Spark Environment (on Master)

```
cd /opt/spark/conf
cp spark-env.sh.template spark-env.sh
nano spark-env.sh
```

Add these variables (adjust IPs accordingly):

```
export SPARK_MASTER_HOST='your-master-ip'
export JAVA_HOME='/usr/lib/jvm/java-11-openjdk-amd64'
```

8. Define Workers (on Master)

```
cp workers.template workers
nano workers
```

Add the IP addresses or hostnames of your worker nodes:

```
worker1-ip
worker2-ip
worker3-ip
```

You can add as many worker IPs as you have VMs. Just ensure you have completed the prerequisite steps (Java, SSH keys) on every node you add.

Starting the Cluster

9. Start Spark Master

On the **Master** node:

```
start-master.sh
```

You can now access the Spark Web UI at <http://your-master-ip:8080>.

10. Start Spark Workers

You can start workers from the Master:

```
start-workers.sh
```

Alternatively, manually start a worker on each **Worker** node:

```
start-worker.sh spark://your-master-ip:7077
```

Verification

11. Test with Spark Shell

Run a simple calculation to ensure the cluster is working:

```
spark-shell --master spark://your-master-ip:7077
```

12. Stop the Cluster

On the **Master** node:

```
stop-all.sh
```